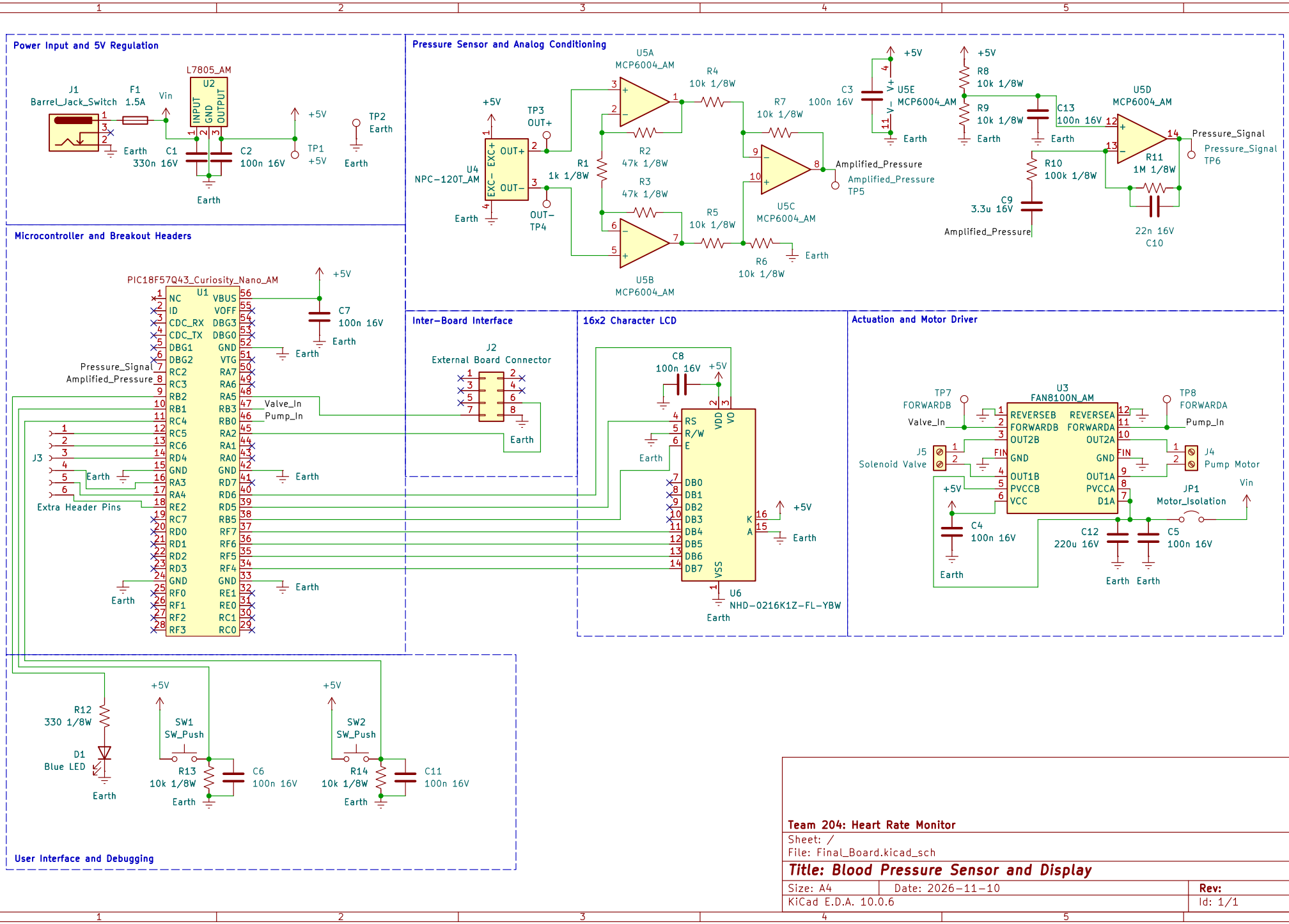

Power Input and 5V Regulation
J1
Barrel_Jack_Switch
F1
1.5A
Vin
L7805_AM
U2
INPUT
GND
OUTPUT
+5V
TP2
Earth
Earth
C1
330n 16V
C2
100n 16V
TP1
+5V
Earth
Pressure Sensor and Analog Conditioning
U5A
MCP6004_AM
R4
10k 1/8W
+5V
TP3
OUT+
U4
NPC-120T_AM
EXC+
EXC-
OUT+
OUT-
R1
1k 1/8W
R2
47k 1/8W
R3
47k 1/8W
R5
10k 1/8W
R6
10k 1/8W
R7
10k 1/8W
U5B
MCP6004_AM
U5C
MCP6004_AM
C3
100n 16V
U5E
MCP6004_AM
Amplified_Pressure
TP5
OUT-
TP4
Earth
R8
10k 1/8W
R9
10k 1/8W
C13
100n 16V
U5D
MCP6004_AM
R10
100k 1/8W
R11
1M 1/8W
C9
3.3u 16V
22n 16V
C10
Pressure_Signal
TP6
Microcontroller and Breakout Headers
PIC18F57Q43_Curiosity_Nano_AM
U1
C7
100n 16V
Pressure_Signal
Amplified_Pressure
Valve_In
Pump_In
J3
Extra Header Pins
Inter-Board Interface
J2
External Board Connector
16x2 Character LCD
C8
100n 16V
RS
R/W
E
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
VDD
VO
VSS
U6
NHD-0216K1Z-FL-YBW
Actuation and Motor Driver
TP7
FORWARDB
TP8
FORWARDA
U3
FAN8100N_AM
REVERSEB
FORWARDB
OUT2B
GND
OUT1B
PVCCB
VCC
REVERSEA
FORWARDA
OUT2A
GND
OUT1A
PVCCA
D1A
FIN
J5
Solenoid Valve
J4
Pump Motor
JP1
Motor_Isolation
C4
100n 16V
C12
220u 16V
C5
100n 16V
User Interface and Debugging
R12
330 1/8W
D1
Blue LED
SW1
SW_Push
R13
10k 1/8W
C6
100n 16V
SW2
SW_Push
R14
10k 1/8W
C11
100n 16V
Team 204: Heart Rate Monitor
Sheet: /
File: Final_Board.kicad_sch
Title: Blood Pressure Sensor and Display
Size: A4
Date: 2026-11-10
Rev:
KiCad E.D.A. 10.0.6
Id: 1/1